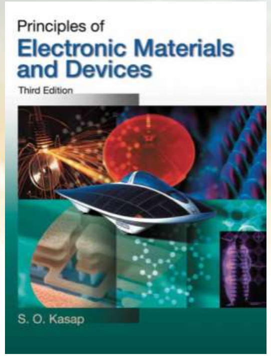
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》
Solid-State Electronics

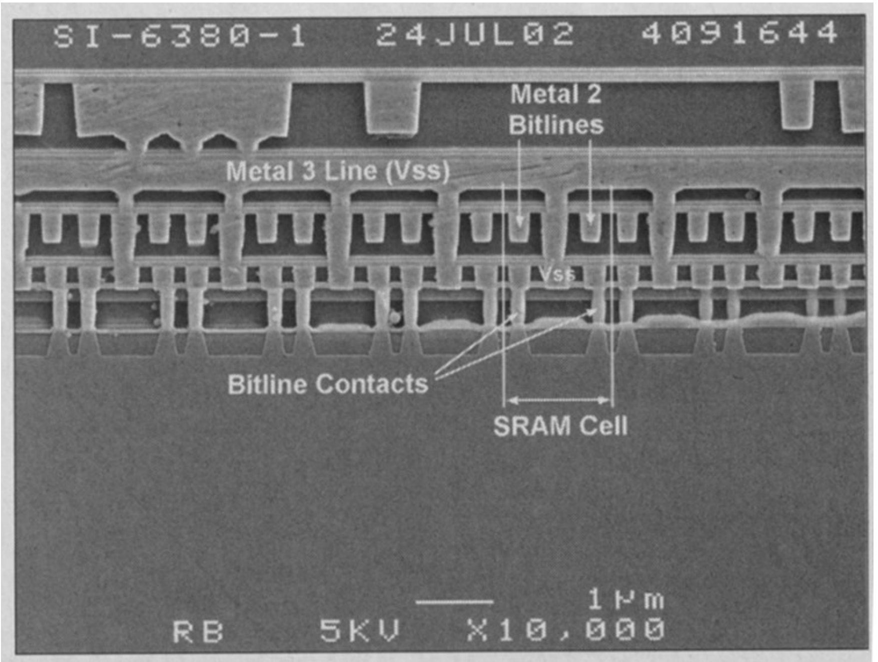


Ch2-1
Electrical & Thermal Conduction in Solids

Dr. Rui CHEN (陈锐)
Email: chenr@sustech.edu.cn
Office: Room 428 at College of Engineering South Tower
Tel: 0755-88018563

Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

Electrical and thermal conduction in solids
固体中的电导和热导

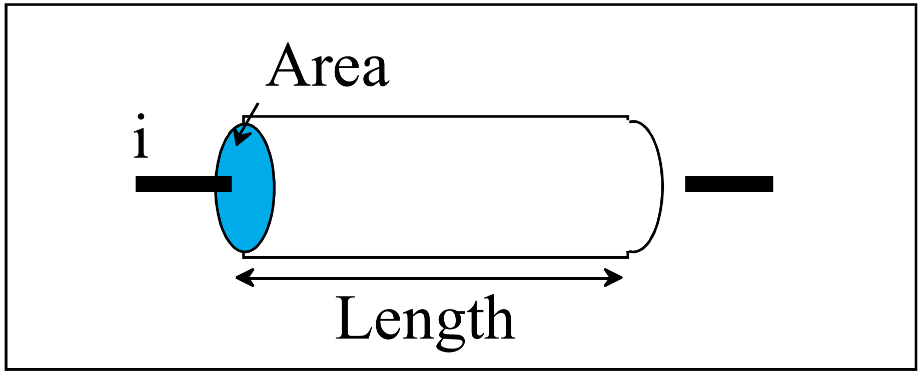


5. A SEM cross section of the interconnect in an SRAM cell.

Metal interconnects are used in microelectronics to wire the devices within the chip, the intergraded circuit. Multilevel interconnects are used for implementing the necessary interconnections.

Resistivity & Conductivity

For a uniform conductor, we have the following relationship



Ohm's Law

$$I = \frac{V}{R}$$

$$R = \rho \frac{L}{S} \qquad \rho = \frac{RS}{L} \qquad \sigma = \frac{1}{\rho}$$

Differential form of Ohm's law $\quad J = \sigma E$

Resistivities of Real Materials

Compound	Resistivity (Ω-cm)	Compound	Resistivity (Ω-cm)
Ca	3.9×10^{-6}	Si	~ 0.1
Ti	42×10^{-6}	Ge	~ 0.05
Mn	185×10^{-6}	ReO_3	36×10^{-6}
Zn	5.9×10^{-6}	Fe_3O_4	52×10^{-6}
Cu	1.7×10^{-6}	TiO_2	9×10^{4}
Ag	1.6×10^{-6}	ZrO_2	1×10^{9}
Pb	21×10^{-6}	Al_2O_3	1×10^{19}

Common Wire and Cable Materials

- **Pure metal**: such as copper, aluminum, iron, etc.

➢ Low resistivity: from small to large, silver (Ag), copper (Cu), gold (Au), aluminum (Al), nano (Na), molybdenum (Mo), tungsten (W), zinc (Zn), nickel (Ni), iron (Fe), platinum (Pt), tin (Sn), lead (Pb), and the like.

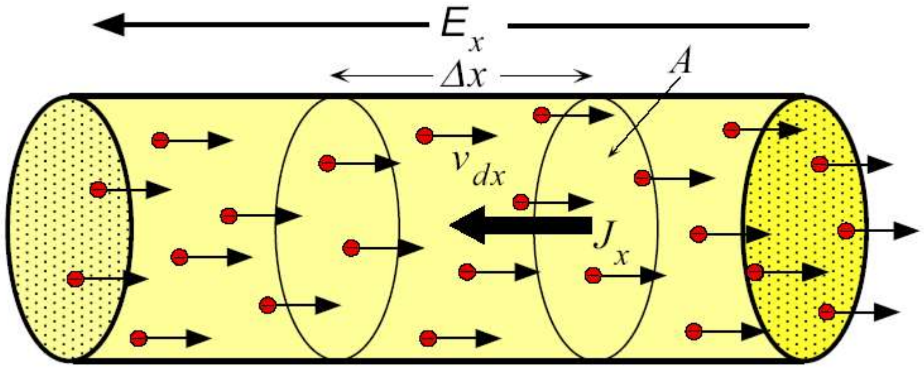
- **Alloys**: such as copper alloy, silver copper, cadmium copper, chromium copper, bismuth copper, zirconium copper, etc.; aluminum alloy, aluminum magnesium silicon, aluminum magnesium, aluminum magnesium iron , aluminum zirconium, etc.

➢ Good mechanical properties or anti-corrosion

2.1 Classical theory: Drude Model 德鲁德模型

Electric current density （电流密度） J： The net amount of charge flowing across a unit area per unit time.

$$J = \frac{\Delta q}{A \Delta t}$$



$E_x \Rightarrow$
Coulombic force eE_x

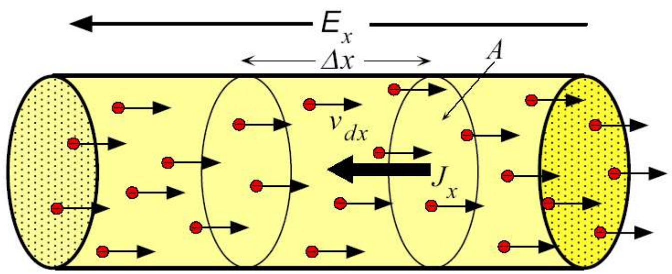
Drift of electrons in a conductor in the presence of an applied electric field. Electrons drift with an average velocity v_{dx} in the x-direction. (E_x is the electric field)

Drift velocity （漂移速率）$v_{dx}(t)$: the average velocity of the electrons in the x direction at time t.

平均瞬时速度

$$v_{dx} = \frac{1}{N}\left[v_{x1} + v_{x2} + v_{x3} + \ldots + v_N\right]$$

N: the number of conduction electrons in the metal



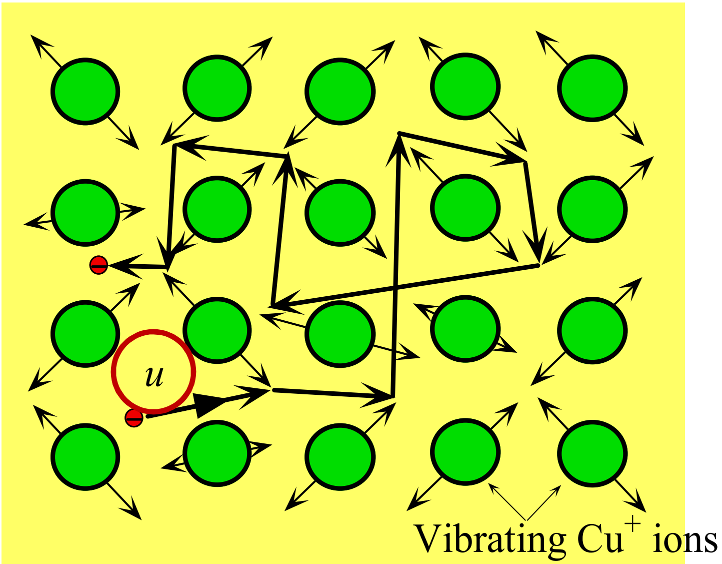
$$\Delta x = v_{dx}\Delta t \qquad n = N/V$$

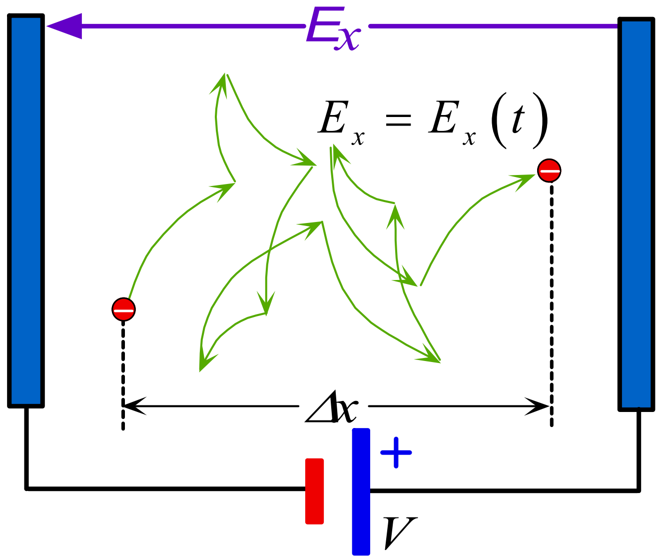
n: the number of electrons per unit volume in the conductor

$$\Delta q = enA\Delta x$$

$$J_x = \frac{\Delta q}{A\Delta t} = \frac{enAv_{dx}\Delta t}{A\Delta t} = env_{dx} \quad \Rightarrow \quad J_x(t) = env_{dx}(t)$$

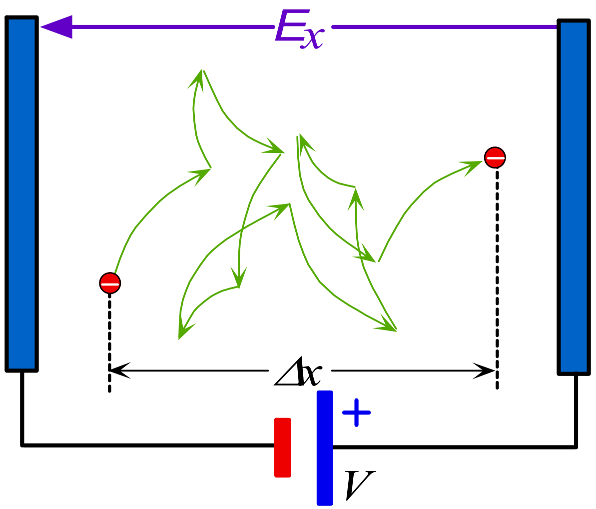
The mean *KE* of the conduction electrons （导带电子）in a metal is primarily determined by the electrostatic interaction （静电） of these electrons with the positive metal ions and also with other.

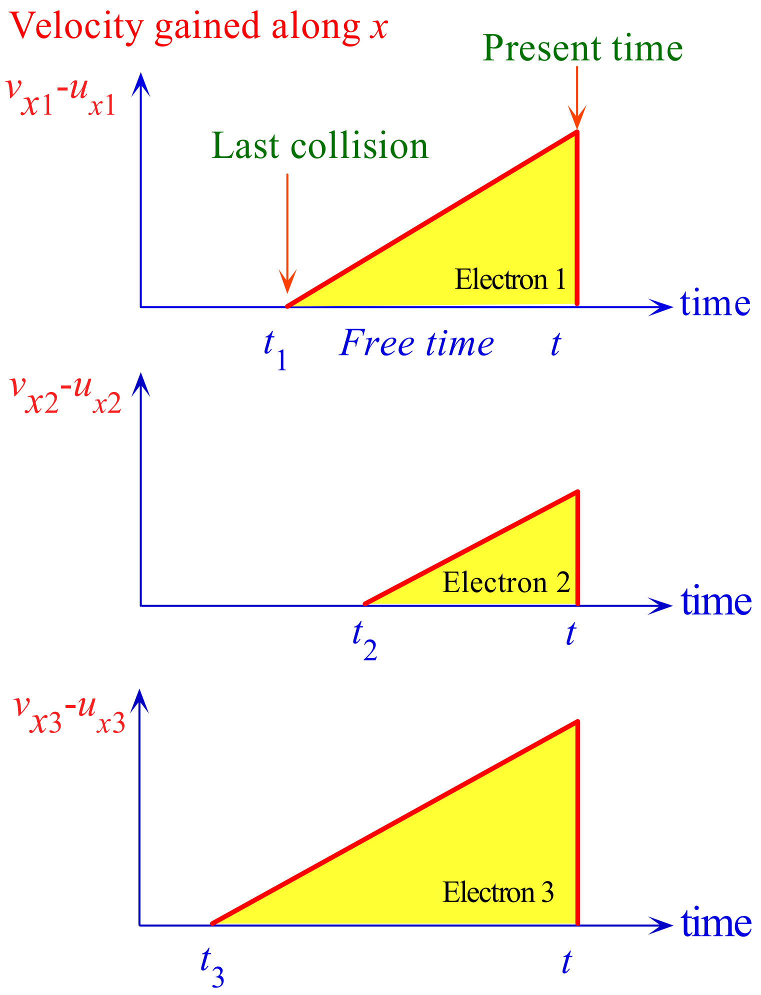




(a) A conduction electron in the electron gas moves about randomly in a metal (with a mean speed u) being frequently and randomly scattered by thermal vibrations of the atoms. In the absence of an applied field there is no net drift in any direction.

(b) In the presence of an applied field, E_x, there is a net drift along the x-direction. This net drift along the force of the field is superimposed on the random motion of the electron. After many scattering events the electron has been displaced by a net distance, Δx, from its initial position toward the positive terminal.





$$v_{xi} = u_{xi} + \frac{eE_x}{m_e}(t - t_i)$$

Velocity gained in the x-direction at time t from the electric field (E_x) for three electrons. There will be N electrons to consider in the metal.

$$v_{xi} = u_{xi} + \frac{eE_x}{m_e}(t - t_i)$$ Initial velocity

$$v_{dx} = \frac{1}{N}\left[v_{x1} + v_{x2} + v_{x3} + \ldots + v_N\right]$$

$$= \frac{eE_x}{m_e}\overline{(t - t_i)}$$

$$\overline{(t - t_i)} = \tau$$

τ : The mean free time, the mean time between collisions.

$$v_{dx} = \frac{eE_x}{m_e}\overline{(t - t_i)} = \frac{e\tau}{m_e}E_x = \mu_d E_x$$

$$\mu_d = \frac{e\tau}{m_e}$$ drift mobility

$$\Rightarrow \quad v_{dx} = \mu_d E_x$$

$$\left.\begin{aligned} & J_x(t) = env_{dx}(t) = en\mu_d E_x \\ & J = \sigma E \end{aligned}\right\} \Rightarrow \quad \sigma = en\mu_d$$

σ: conductivity

Drift mobility is important because it is a widely used electronic parameter in semiconductor device physics. The drift mobility gauges how fast electrons will drift when driven by an applied field. If the electron is not highly scattered, then the mean free time between collisions will be long, the drift mobility will also be large. However, a large drift mobility does not necessarily imply high conductivity, because it also depends on the concentration of conduction electrons.

PROBABILITY OF SCATTERING PER UNIT TIME AND THE MEAN FREE TIME If $1/\tau$ is defined as the mean probability per unit time that an electron is scattered, show that the mean time between collisions is τ.

EXAMPLE 2.1

SOLUTION

Consider an infinitesimally small time interval dt at time t. Let N be the number of unscattered electrons at time t. The probability of scattering during dt is $(1/\tau)\,dt$, and the number of scattered electrons during dt is $N(1/\tau)\,dt$. The change dN in N is thus

$$dN = -N\left(\frac{1}{\tau}\right)dt$$

The negative sign indicates a reduction in N because, as electrons become scattered, N decreases. Integrating this equation, we can find N at any time t, given that at time $t = 0$, N_0 is the total number of unscattered electrons. Therefore,

$$N = N_0\,\exp\left(-\frac{t}{\tau}\right)$$

Unscattered electron concentration

This equation represents the number of unscattered electrons at time t. It reflects an exponential decay law for the number of unscattered electrons. The **mean free time** $\bar{t}$ can be calculated from the mathematical definition of $\bar{t}$,

$$\bar{t} = \frac{\int_0^\infty tN\,dt}{\int_0^\infty N\,dt} = \tau$$

Mean free time

where we have used $N = N_0\exp(-t/\tau)$. Clearly, $1/\tau$ is the mean probability of scattering per unit time.

ELECTRON DRIFT MOBILITY IN METALS Calculate the drift mobility and the mean scattering time of conduction electrons in copper at room temperature, given that the conductivity of copper is $5.9 \times 10^5\ \Omega^{-1}\ \text{cm}^{-1}$. The density of copper is $8.96\ \text{g cm}^{-3}$ and its atomic mass is $63.5\ \text{g mol}^{-1}$.

EXAMPLE 2.2

SOLUTION

We can calculate μ_d from $\sigma = en\mu_d$ because we already know the conductivity σ. The number of free electrons n per unit volume can be taken as equal to the number of Cu atoms per unit volume, if we assume that each Cu atom donates one electron to the conduction electron gas in the metal. One mole of copper has N_A (6.02×10^{23}) atoms and a mass of 63.5 g. Therefore, the number of copper atoms per unit volume is

$$n = \frac{dN_A}{M_{\text{at}}}$$

where d = density = $8.96\ \text{g cm}^{-3}$, and M_{at} = atomic mass = 63.5 (g mol^{-1}). Substituting for d, N_A, and M_{at}, we find $n = 8.5 \times 10^{22}$ electrons cm^{-3}.

The electron drift mobility is therefore

$$\mu_d = \frac{\sigma}{en} = \frac{5.9 \times 10^5\ \Omega^{-1}\ \text{cm}^{-1}}{[(1.6 \times 10^{-19}\ \text{C})(8.5 \times 10^{22}\ \text{cm}^{-3})]}$$

$$= 43.4\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$$

From the drift mobility we can calculate the mean free time τ between collisions by using Equation 2.5,

$$\tau = \frac{\mu_d m_e}{e} = \frac{(43.4 \times 10^{-4}\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1})(9.1 \times 10^{-31}\ \text{kg})}{1.6 \times 10^{-19}\ \text{C}} = 2.5 \times 10^{-14}\ \text{s}$$

Note that the mean speed u of the conduction electrons is about $1.5 \times 10^6\ \text{m s}^{-1}$, so that their mean free path is about 37 nm.

EXAMPLE 2.3 **DRIFT VELOCITY AND MEAN SPEED** What is the applied electric field that will impose a drift velocity equal to 0.1 percent of the mean speed $u(\sim 10^6 \text{ m s}^{-1})$ of conduction electrons in copper? What is the corresponding current density and current through a Cu wire of diameter 1 mm?

SOLUTION

The drift velocity of the conduction electrons is $v_{dx} = \mu_d \mathcal{E}_x$, where μ_d is the drift mobility, which for copper is 43.4 cm^2 V^{-1} s^{-1} (see Example 2.2). With $v_{dx} = 0.001u = 10^3 \text{ m s}^{-1}$, we have

$$\mathcal{E}_x = \frac{v_{dx}}{\mu_d} = \frac{10^3 \text{ m s}^{-1}}{43.4 \times 10^{-4} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}} = 2.3 \times 10^5 \text{ V m}^{-1} \quad \text{or} \quad 230 \text{ kV m}^{-1}$$

This is an unattainably large electric field in a metal. Given the conductivity σ of copper, the equivalent current density is

$$J_x = \sigma \mathcal{E}_x = (5.9 \times 10^7 \ \Omega^{-1} \text{ m}^{-1})(2.3 \times 10^5 \text{ V m}^{-1})$$
$$= 1.4 \times 10^{13} \text{ A m}^{-2} \quad \text{or} \quad 1.4 \times 10^7 \text{ A mm}^{-2}$$

This means a current of 1.1×10^7 A through a 1 mm diameter wire! It is clear from this example that for all practical purposes, even under the highest working currents and voltages, the drift velocity is much smaller than the mean speed of the electrons. Consequently, when an electric field is applied to a conductor, for all practical purposes, the mean speed is unaffected.

Drift of Carrier （载流子漂移）

Mobility （迁移率）

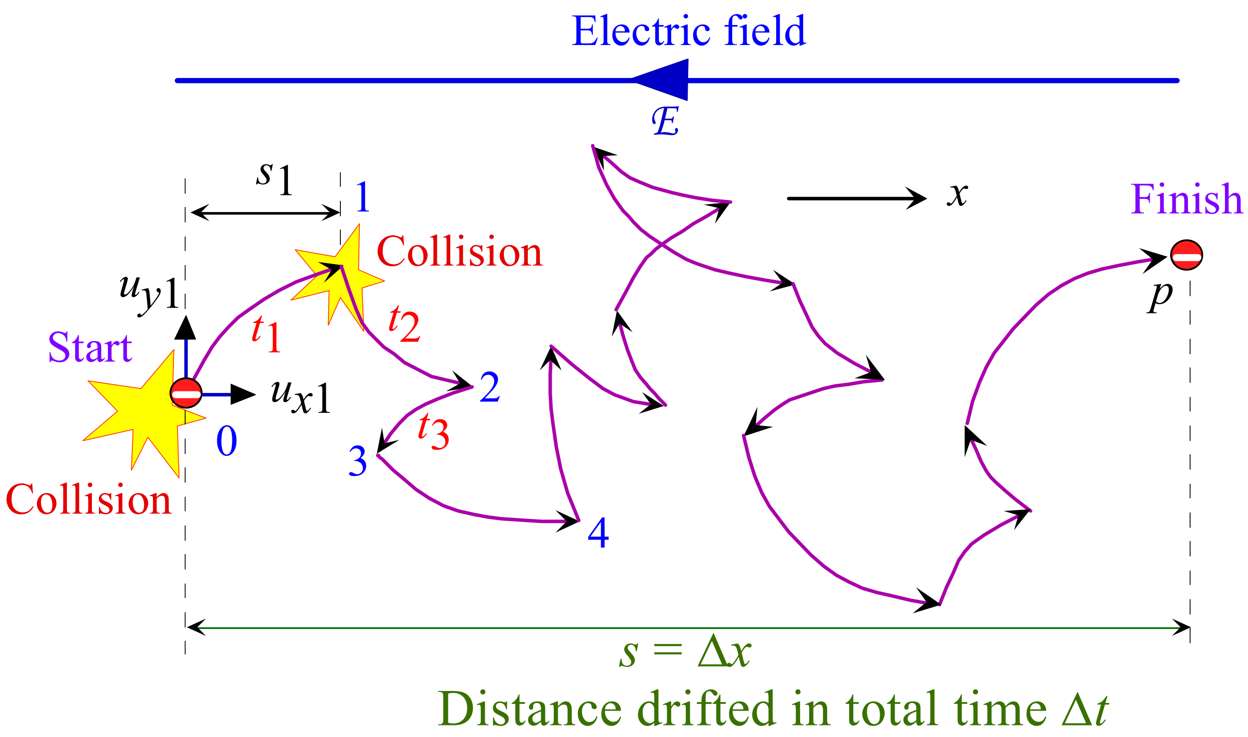
Mobility is a physical quantity used to describe the movement of carriers in a semiconductor under a unit electric field. It is an important parameter describing the phenomenon of carrier transport and it is a very important basic concept in semiconductor.

Definition $$\mu = \frac{q\tau_c}{m}$$ unit: cm²/(V·s)

Since carriers have electrons and holes, mobility also has electron mobility and hole mobility, namely:

Electron Mobility $$\mu_e = \frac{e\tau_c}{m_e}$$

Hole Mobility $$\mu_p = \frac{q\tau_c}{m_p}$$



The motion of a single electron in the presence of an electric field E. During a time interval t_i, the electron traverses a distance s_i along x. After p collisions, it has drifted a distance $s = \Delta x$.

The distance S_1 covered in the x direction during the free time t_1 will be

$$S_1 = u_{x1} t_1 + \frac{1}{2}\left(\frac{eE_x}{m_e}\right) t_1^2$$

$$S = S_1 + S_1 + S_1 + \ldots + S_p = \frac{1}{2}\left(\frac{eE_x}{m_e}\right) p\overline{t^2}$$

$$\overline{t^2} = \frac{1}{p}\left(t_1^2 + t_2^2 + t_3^2 + \ldots + t_p^2\right)$$

$\overline{t^2}$ mean square free time（均方自由时间）

$$\tau = \frac{1}{p}\left(t_1 + t_2 + t_3 + \ldots + t_p\right)$$

τ: mean free time（平均自由时间）

$$\overline{t^2} = 2\left(\bar{t}\right)^2 = 2\tau^2$$

$$\Rightarrow \quad S = \left(\frac{eE_x}{m_e}\right)\left(p\tau^2\right)$$

$$\Rightarrow \quad v_{dx} = \frac{\Delta x}{\Delta t} = \frac{S}{p\tau} = \frac{e\tau}{m_e} E_x$$

$$l = u\tau$$

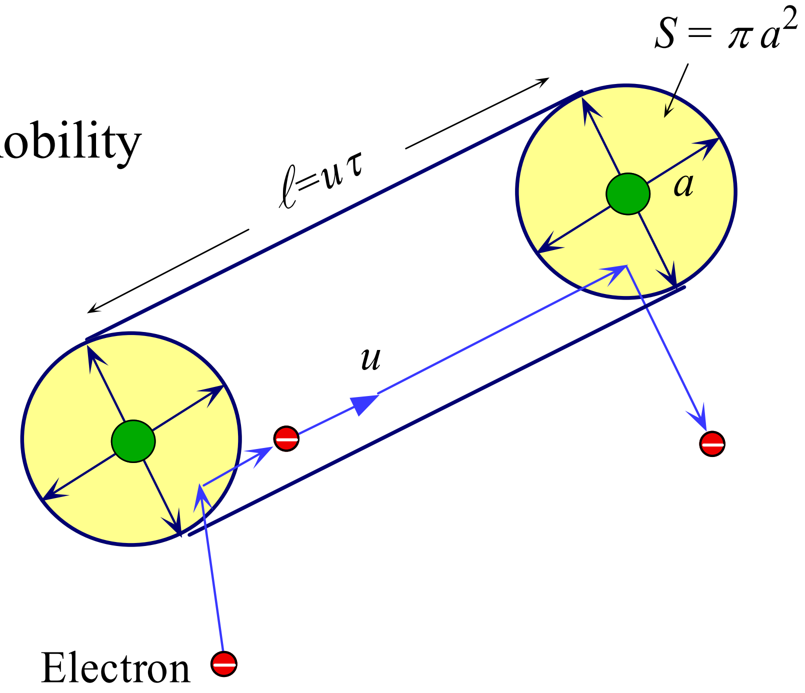
mean free path（平均自由程）

2.2 Metallic Conductive Materials

- Factors affecting metal resistivity
  - ✓ Impurities and defects: generally, resistivity of alloy is higher than pure metal
  - ✓ **Temperature**
  - ✓ Pressure $\rho(p) = \rho_0(1 + \alpha p)$

Temperature dependence of resistivity: Ideal pure metals

$$\mu_d = \frac{e\tau}{m_e}$$ drift mobility



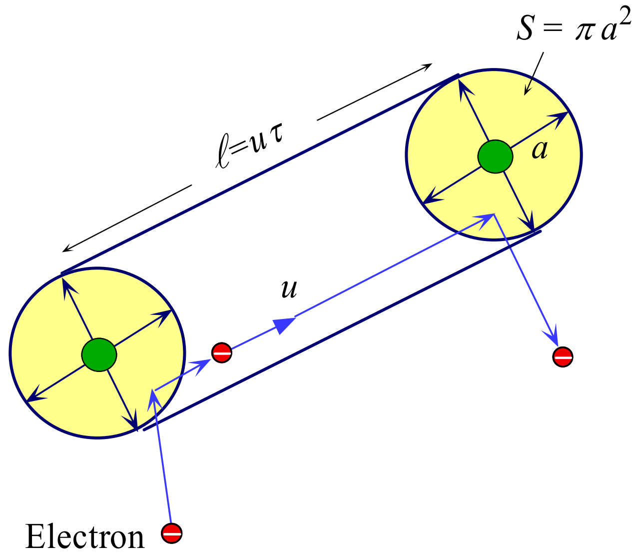
Scattering of an electron from the thermal vibrations of the atoms. The electron travels a mean distance $\ell = u\ \tau$ between collisions. Since the scattering cross sectional area is S, in the volume $S\ell$ there must be at least one scatterer, $N_S(Su\tau) = 1$.

N_s: concentration of scattering centers
（散射中心浓度）

Temperature dependence of resistivity: Ideal pure metals

$$l = u\tau$$
$$N_s(Su\tau) = 1 \quad \Longrightarrow \quad \tau = \frac{1}{SuN_s}$$

N_s: the concentration of scattering centers.



$$\frac{1}{4}Ma^2\omega^2 \approx \frac{1}{2}kT \quad \Rightarrow \quad a^2 \propto T$$
$$N_s(Su\tau) = 1 \quad \Rightarrow \quad \tau \propto \frac{1}{\pi a^2} \propto \frac{1}{T}$$
$$\Longrightarrow \quad \tau = \frac{C}{T}$$

$$\Longrightarrow \quad \rho_T = \frac{1}{\sigma_T} = \frac{1}{en\mu_d} = \frac{m_e T}{e^2 nC}$$

$$\rho_T \propto T$$

EXAMPLE 2.6 **DRIFT MOBILITY AND RESISTIVITY DUE TO LATTICE VIBRATIONS** Given that the mean speed of conduction electrons in copper is 1.5×10^6 m s^{-1} and the frequency of vibration of the copper atoms at room temperature is about 4×10^{12} s^{-1}, estimate the drift mobility of electrons and the conductivity of copper. The density d of copper is 8.96 g cm^{-3} and the atomic mass M_{at} is 63.56 g mol^{-1}.

SOLUTION

The method for calculating the drift mobility and hence the conductivity is based on evaluating the mean free time τ via Equation 2.11, that is, $\tau = 1/SuN_s$. Since τ is due to scattering from atomic vibrations, N_s is the atomic concentration,

$$N_s = \frac{dN_A}{M_{at}} = \frac{(8.96 \times 10^3 \text{ kg m}^{-3})(6.02 \times 10^{23} \text{ mol}^{-1})}{63.56 \times 10^{-3} \text{ kg mol}^{-1}}$$

$$= 8.5 \times 10^{28} \text{ m}^{-3}$$

The cross-sectional area $S = \pi a^2$ depends on the amplitude a of the thermal vibrations as shown in Figure 2.5. The average kinetic energy KE_{av} associated with a vibrating mass M attached to a spring is given by $KE_{av} = \frac{1}{4}Ma^2\omega^2$, where ω is the angular frequency of the vibration ($\omega = 2\pi 4 \times 10^{12}$ rad s^{-1}). Applying this equation to the vibrating atom and equating the average kinetic energy KE_{av} to $\frac{1}{2}kT$, by virtue of equipartition of energy theorem, we have $a^2 = 2kT/M\omega^2$ and thus

$$S = \pi a^2 = \frac{2\pi kT}{M\omega^2} = \frac{2\pi(1.38 \times 10^{-23} \text{ J K}^{-1})(300 \text{ K})}{\left(\dfrac{63.56 \times 10^{-3} \text{ kg mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}}\right)(2\pi \times 4 \times 10^{12} \text{ rad s}^{-1})^2}$$

$$= 3.9 \times 10^{-22} \text{ m}^2$$

Therefore,

$$\tau = \frac{1}{SuN_s} = \frac{1}{(3.9 \times 10^{-22}\ \mathrm{m^2})(1.5 \times 10^6\ \mathrm{m\ s^{-1}})(8.5 \times 10^{28}\ \mathrm{m^{-3}})}$$

$$= 2.0 \times 10^{-14}\ \mathrm{s}$$

The drift mobility is

$$\mu_d = \frac{e\tau}{m_e} = \frac{(1.6 \times 10^{-19}\ \mathrm{C})(2.0 \times 10^{-14}\ \mathrm{s})}{(9.1 \times 10^{-31}\ \mathrm{kg})}$$

$$= 3.5 \times 10^{-3}\ \mathrm{m^2\ V^{-1}\ s^{-1}} = 35\ \mathrm{cm^2\ V^{-1}\ s^{-1}}$$

The conductivity is then

$$\sigma = en\mu_d = (1.6 \times 10^{-19}\ \mathrm{C})(8.5 \times 10^{22}\ \mathrm{cm^{-3}})(35\ \mathrm{cm^2\ V^{-1}\ s^{-1}})$$

$$= 4.8 \times 10^{5}\ \Omega^{-1}\ \mathrm{cm^{-1}}$$

The experimentally measured value for the conductivity is $5.9 \times 10^5\ \Omega^{-1}\ \mathrm{cm^{-1}}$, so our crude calculation based on Equation 2.11 is actually only 18 percent lower, which is not bad for an estimate. (As we might have surmised, the agreement is brought about by using reasonable values for the mean speed u and the atomic vibrational frequency ω. These values were taken from quantum mechanical calculations, so our evaluation for τ was not truly based on classical concepts.)